

Ranked Research Roadmap & Kickoff

A re-ranked research program after the synthetic geometry phases. This report argues for a shift from manifold-first work toward blocked-valence, settings-twist, and causal-decomposition work, then shows the live DB audit and kickoff manifests that make that shift executable immediately.

DATE	LIVE DECISIONS	TOP TRACK	KICKOFF BATCH
20 March 2026	121,352	Blocked valence + settings twist	154 rows

MAIN READ

The best research path is no longer manifold-first.

The highest-value program now is: determine what the model wants to do, determine what stops it, and determine whether settings reweight that preference or merely gate execution. Geometry still matters, but as a simplifying lens for a behavioral and causal story rather than as the main scientific target.

BLOCKED POOL	POLICY-TENSION POOL	BLOCKED KICKOFF	SETTINGS KICKOFF
59,468 live candidate rows	30,115 live candidate rows	34 unique-vault starter batch	120 72 observe + 48 refs

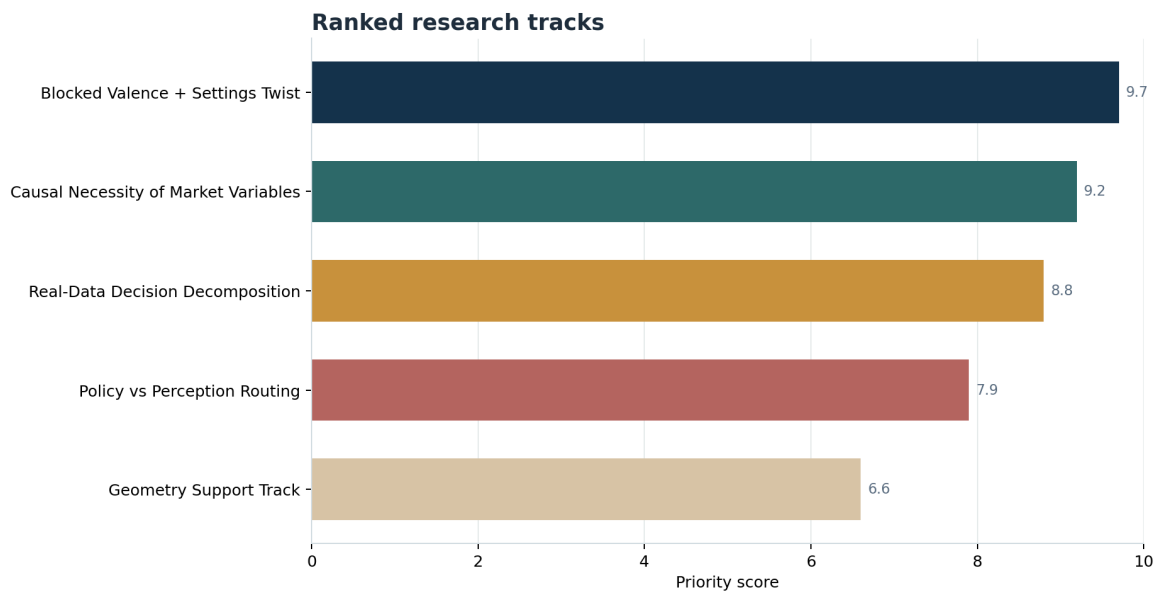
Why The Program Should Shift

The current evidence stack no longer supports treating manifold discovery as the center of gravity.

- Decision-structure work showed **where** action-relevant signal lives, but not what intermediate quantities the model actually computes.
- Synthetic scalar and coupled geometry work showed some low-dimensional order, but not a single clean market manifold.
- The strongest current unresolved questions are behavioral and causal:
 - is observe neutral or blocked?
 - are settings late gates or true reinterpretation operators?
 - what variables are actually necessary for the model to choose an action?

This report therefore re-ranks the research tracks around those questions.

Ranked Research Tracks



Priority scores for the next research tracks. The top-ranked work is blocked valence plus settings twist, followed by causal necessity of the strongest market variables.

Rank	Track	Score	Why it belongs here
1	Blocked valence + settings twist	9.7	Shortest path from localization to mechanism; creates the missing labels for asset valence and directly tests whether settings reweight preference or only gate execution.
2	Causal necessity of market variables	9.2	The synthetic work has already surfaced the strongest candidate variables; the next gain comes from patching and ablation rather than more passive decoding.
3	Real-data decision decomposition	8.8	Reconnects synthetic findings to real DX-terminal prompts and separates true neutral observe cases from blocked-latent-sentiment cases.
4	Policy vs perception routing	7.9	Potentially cleaner mechanistic handle if experts split along market parsing versus policy handling.
5	Geometry support track	6.6	Still useful, but should simplify causal stories rather than define the main agenda.

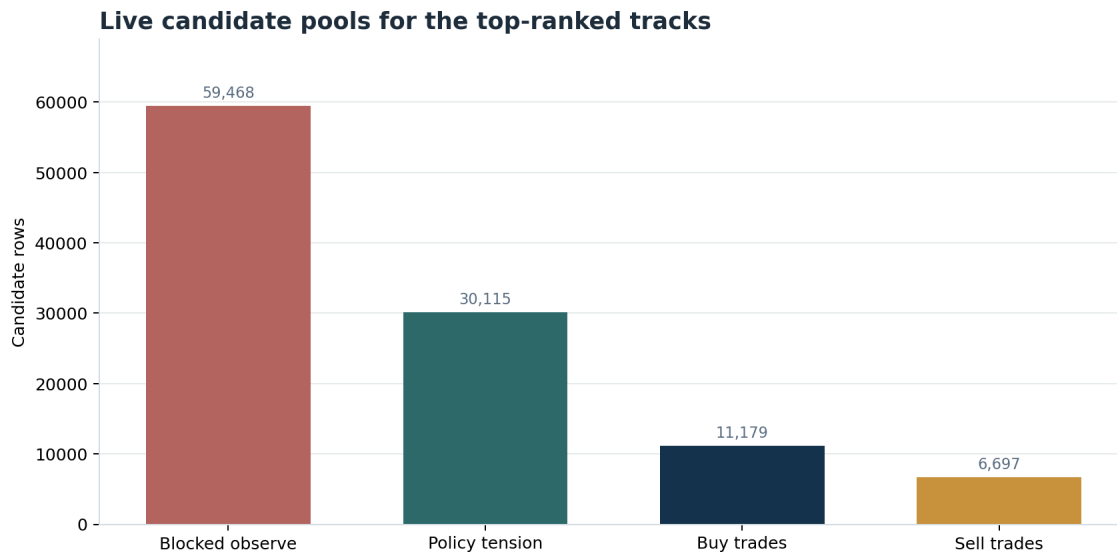
Top Hypotheses

The most interesting hypotheses to explore next are:

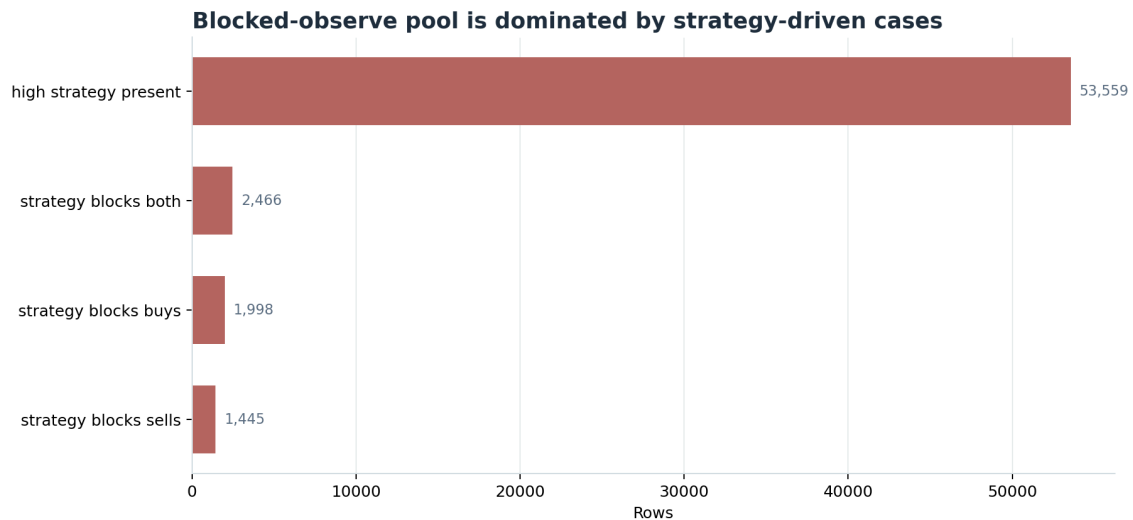
- **Early preference, late permission.** Market preference forms before legality and affordance gating.
- **Blocked observe hides valence.** Many observe cases are actually bullish or bearish states that cannot be executed.
- **Settings reweight, not rewrite.** Settings mostly rotate or sharpen existing preference structure rather than creating preference from scratch.
- **Flow matters jointly.** Momentum $\times$ flow is a stronger candidate mechanism than flow alone.
- **Participation modulates confidence.** Participation shapes whether momentum is trusted rather than acting as a primary preference axis.
- **Concentration is policy-facing.** Concentration matters more through risk or policy pathways than as a clean early perceptual variable.

Live Candidate Audit

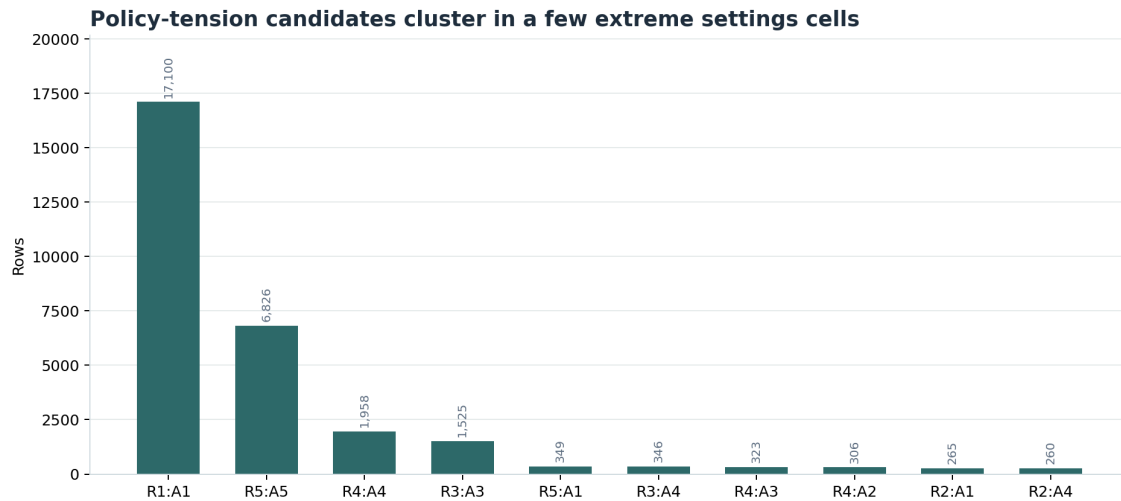
The live Neon pool is already large enough to support the top-ranked program.



The raw pool is not the bottleneck anymore. There are already tens of thousands of blocked-observe and policy-tension examples available for the next research phase.



The blocked-observe pool is huge but structurally narrow: most cases are strategy-driven, which means blocked-valence reruns should focus on deconstraint and strategy-removal variants rather than assuming a broad natural mix.



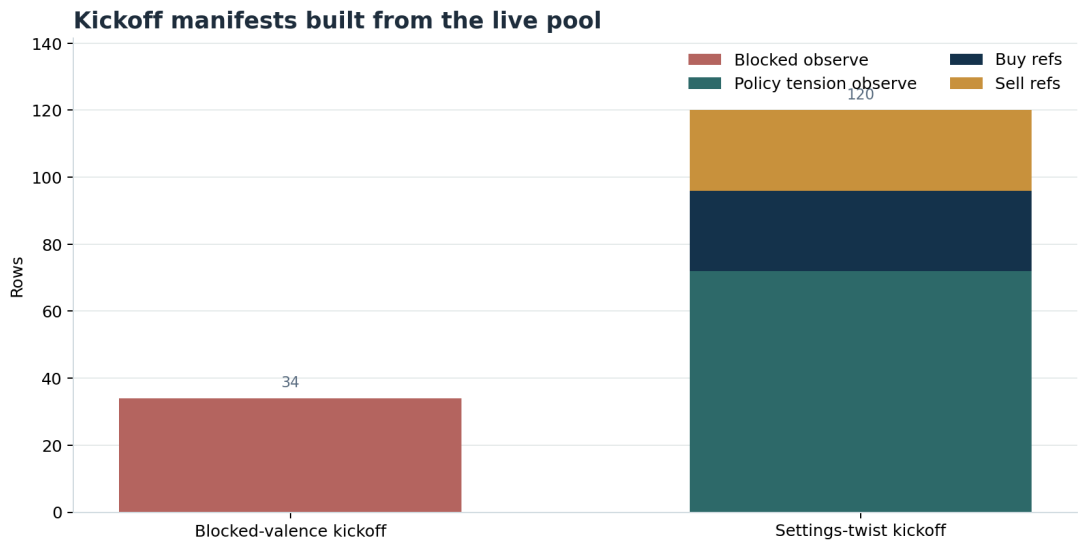
Policy-tension cases cluster heavily in a few extreme settings regimes, especially R1:A1 and R5:A5. That is useful rather than bad: it means the first settings-twist pass can focus on high-contrast cells instead of diffuse mid-range settings.

Most important audit reads:

- The blocked pool is large (59,468) but dominated by high_strategy_present cases (53,559), with smaller secondary blocks from explicit buy/sell restrictions.
- The policy-tension pool is also large (30,115) and highly actionable: 28,500 cases still have both buy and sell available.
- Natural settings diversity is concentrated, not uniform. Extreme settings cells dominate the pool, which is exactly what the first twist experiments should exploit.

Kickoff Manifests

The kickoff is already underway, and the live audit changed the operational design: one combined manifest was the wrong abstraction because observe-heavy rows were competing with trade-reference rows for the same vault budget. The correct shape is two manifests.



The kickoff now splits into two manifests: a blocked-valence starter batch and a larger settings-twist batch with explicit trade references.

Manifest	Unique vaults	Use
Blocked valence kickoff	34	Deconstraint / strategy-removal reruns on blocked observe cases.
Settings twist kickoff	102	72 policy-tension observe cases plus 24 buy and 24 sell reference trades.

Interpretation:

- The blocked-valence starter is intentionally smaller because the highest-priority blocked pool is structurally concentrated.
- The settings-twist starter is larger because the candidate pool is both large and highly actionable.
- This is enough to begin the top-ranked research track without waiting for more raw data or more geometry work.

Immediate Next Experiments

The next concrete experiments should be:

1. rerun the blocked-valence kickoff batch under deconstraint or strategy removal
2. rerun the settings-twist kickoff batch under high-contrast settings rewrites

3. compare `row_mean_i`, `active_settings_eos`, `constraints_eos`, and `last_token`
4. use the rerun outcomes to build the first blocked-bullish and blocked-bearish labels

Only after those experiments are in place should the program widen again into broader geometry or routing work.

What To Deprioritize

- broad manifold search without a behavioral decomposition
- more generic buy/sell probes without blocked-valence labels
- treating observe as uniformly neutral
- expanding synthetic geometry breadth before testing the top behavioral hypotheses

Conclusion

The live data and the current result stack now point in the same direction:

- the research program should be organized around **decision decomposition**, not just geometry
- the best immediate target is **blocked valence plus settings twist**
- the kickoff is already materially underway with live candidate audits and runnable starter manifests

This is a better place to spend the next iteration than another round of broad manifold hunting.

Ranked Research Roadmap & Kickoff — 20 March 2026. Based on live Neon audit results plus prior decision-structure and synthetic geometry findings.